◀ **Figure 34.43 Verreaux's sifakas (*Propithecus verreauxi*), a type of lemur.**

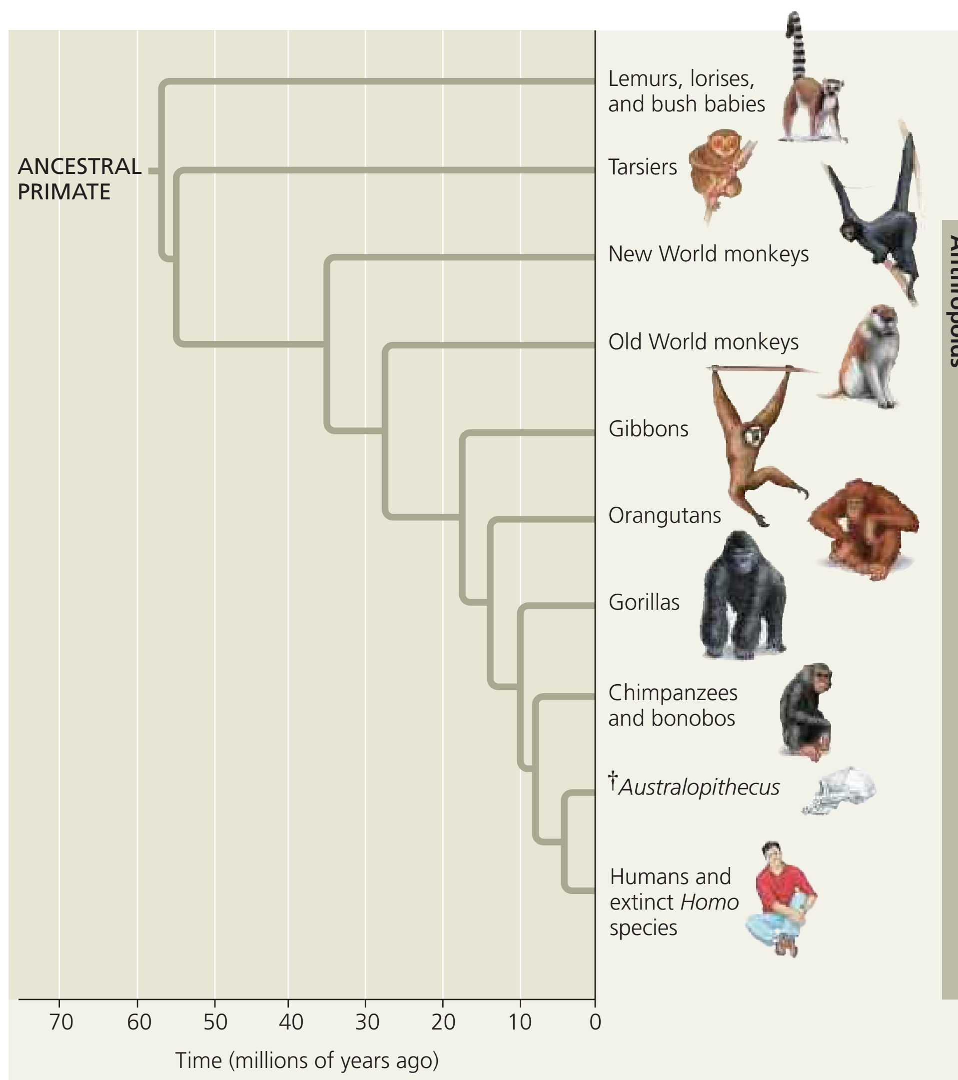
reached South America from Africa by rafting on logs or other debris. What is certain is that New World monkeys and Old World monkeys underwent separate adaptive radiations during their many millions of years of separation **(Figure 34.45)**. All species of New World monkeys are arboreal, whereas Old World monkeys include ground-dwelling as well as arboreal species. Most monkeys in both groups are diurnal (active during the day) and usually live in bands held together by social behavior.

The other group of anthropoids consists of primates informally called apes **(Figure 34.46)**. The ape group includes four genera of gibbons, along with the so-called "great apes," the genera *Pongo* (orangutans), *Gorilla* (gorillas), *Pan* (chimpanzees and bonobos), and *Homo* (humans). The apes diverged from Old World monkeys

ancestors. Arboreal maneuvering also requires excellent eye-hand coordination. The overlapping visual fields of the two forward-facing eyes enhance depth perception, an obvious advantage when brachiating (traveling by swinging from branch to branch in trees).

Living Primates There are three main groups of living primates: (1) the lemurs of Madagascar **(Figure 34.43)** and the lorises and bush babies of tropical Africa and southern Asia; (2) the tarsiers, which live in southeastern Asia; and (3) the **anthropoids**, which include monkeys and apes and are found worldwide. The first group—lemurs, lorises, and bush babies—probably resemble early arboreal primates. The oldest known tarsier fossils date to 55 million years ago; along with DNA evidence, these fossils indicate that tarsiers are more closely related to anthropoids than to the lemur group **(Figure 34.44)**.

You can see in Figure 34.44 that monkeys do not form a clade but rather consist of two groups, the New and Old World monkeys. Both of these groups are thought to have originated in Africa or Asia. The fossil record indicates that New World monkeys first colonized South America roughly 25 million years ago. By that time, South America and Africa had drifted apart, and monkeys may have



▲ **Figure 34.44 A phylogenetic tree of primates.** The fossil record indicates that the lineage leading to monkeys and apes diverged from other primates 55 million years ago. The lineage leading to humans and *Australopithecus* branched off from other apes about 8 million years ago; the dagger symbol (†) denotes an extinct group.

VISUAL SKILLS *Is the phylogeny shown here consistent with the idea that humans evolved from chimpanzees? Explain.*